SAARTHAK KHANDELWAL

AI & SOFTWARE ENGINEER

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EDUCATION

2026

VELLORE INSTITUTE OF TECHNOLOGY, VELLORE

- B.Tech Computer Science Engineering
- GPA- 7.82/10

2022

GREENWOOD HIGH, BANGALORE

- ICSE/ISC | X - 98%, XII - 92%

SKILLS

Programming Languages:
- Python, Java, C, C++, PHP, JavaScript

Frameworks & Tools:
- FastAPI, SentenceTransformers, FAISS, OpenAI API, Gemini API, Anthropic API, Qdrant, LangChain, HuggingFace, AutoGluon, PennyLane, Qiskit, REST APIs

Cloud:
- Azure (basic), AWS (basic)

Software:
- VS Code, Blue J, Power BI, Unreal Engine, Unity

Languages:
- English (Fluent), Hindi (Fluent), German (Basic), French (Basic)

PROJECTS

Neuroevolutionary Procedural Level Generator (Personal Project) — 2026
- Built a genetic algorithm in Python/NumPy that evolves 2D platformer levels, using a from-scratch neural network trained to learn a fitness function in place of hand-coded rules; validated with property-based testing (Hypothesis).

Autonomous Driving Simulator (Personal Project) — 2025
- Built a browser-based simulator in vanilla JavaScript where a neural network (trained via genetic algorithm) learns to drive using ray-cast sensor inputs, rendered live via Canvas API.

Automated Lecture Notes Agent (Personal Project) — 2026
- Built a Python agent that converts lecture slides and transcripts into LLM-generated notes with diagrams and verified links, auto-published to Obsidian and Notion

ONLINE COURSES

- CS50's Introduction to AI with Python - HarvardX
- Complete Generative AI with LangChain & HuggingFace - Krish Naik, Udemy
- Machine Learning Bootcamp - Devtown
- Complete 2024 Web Development Bootcamp - Angela Yu, Udemy
- Vue.js Fast Crash Course - Edwin Diaz, Udemy
- Complete C# Unity Game Developer 2D - Rick Davidson, Udemy

PROFILE

Postgraduate student pursuing a Master of Science in Computer Science at the University of Sydney, with a background in AI, machine learning, and backend development. Research and industry experience spans quantum NLP, edtech AI, and procedural/generative systems, with ongoing project work in ML-driven simulation and game development.

WORK EXPERIENCE

Calibr.AI (Brainhive Labs Pvt Ltd) — 2025 - 2026

AI Development Intern

- Enhanced the LMS platform with AI-driven modules and optimized REST APIs, improving content delivery and website performance for STEM and child education programs.
- Built a RAG-based pipeline for automated learning content generation, delivering dynamic, curriculum-aligned material for STEM and digital-skills programs.
- Engineered an automated skill tagging system using mpnet base v2 with FAISS-indexed taxonomy search and MMR-based diversity reranking, plus embedding-similarity difficulty classification served via FastAPI.

Vesuvius R&D (VRIPL), Vizag, India — 2025 - 2026

Research Intern

- Conducted predictive analysis on refractory lab trial data under the guidance of Dr. Akshay Bauskar, developing ML models to estimate key material properties.
- Designed a complete AutoGluon-based workflow for data cleaning, feature processing, multi-target regression modeling, and evaluation of refractory material behavior.

Vellore Institute of Technology, Vellore, India — 2025

Research Intern

- Worked under the guidance of Dr. Sathiya Kumar C., Professor, SCOPE.
- Developed a hybrid quantum-classical Transformer combining XLM-RoBERTa with VOC and QRC attention layers using PennyLane and Qiskit.
- Achieved 87% (EN) and ~80% (DE/ES) accuracy on XGLUE, outperforming classical baselines with accuracy-efficiency trade-offs analyzed across Hybrid, VQC, QRC, models using PennyLane and Qiskit.
- Preprint DOI: 10.5281/zenodo.21106178

Infinite Computer Solutions, Bangalore, India — 2024

Software Intern (Turbify)

- Navigated and analyzed a large backend codebase while integrating Apple Pay and implementing encrypted, secure data transfer mechanisms.
- Developed knowledge of encryption protocols and resolved real-time payment integration issues, optimizing backend performance and security.

Yale University, CT, USA — 2023 - 2024

Research Intern

- Worked under the guidance of Prof. Amit Khandelwal, Professor of Global Affairs and Economics.
- Analysed the effects of the US-China Trade War using R.
- Performed regression analysis in Python.